

Szemerédi's Regularity Lemma: Are the irregular pairs necessary?

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1 Introduction

Szemerédi's theorem on the existence of arithmetic progressions in subsets with positive upper density is an acclaimed result that generalized ideas and previous theorems. Nowadays, several different proofs are known, many of which come from different fields of mathematics. This is why this theorem has been considered a “Rosetta Stone” [7] that relates seemingly disconnected fields.

This paper aims to give a deeper insight into Szemerédi's regularity lemma, crucial for the theorem mentioned above, by addressing details regarding (un)necessity the irregular pairs described by the lemma. This paper closely follows [5] by Malliaris and Shelah. Throughout this paper it is assumed that the reader is familiar with the concepts used in the lemma statement. Let us first recall the theorem statement and start building from there.

Theorem 1.1 (Equitable Szemerédi's Regularity Lemma)

For every $\varepsilon > 0$ and m_0 there exists a constant $M = M(\varepsilon)$ such that every finite graph G can be partitioned into k sets $\{V_1, V_2, \dots, V_k\}$ where $m_0 \leq k \leq M$ and:

- **Equipartition:** All the pairs V_i, V_j satisfy: $||V_i| - |V_j|| \leq 1$
- **Regularity:** All but εk^2 of the pairs (V_i, V_j) are ε -regular.

Notice that the regularity lemma does not state that all the pairs are ε -regular. Instead, it allows for up to a ε -fraction of them to be irregular. For quite some time it was uncertain if the existence of these irregular pairs was necessary, until the authors in [6] observed that they were in fact needed for certain graphs and gave the following simple example which will be of crucial importance to us.

Definition 1.1 (Half-graph of length n)

A half-graph of length n is a bipartite graph $H_n = (V = A \cup B, E)$ such that $A = \{a_0, a_1, \dots, a_{n-1}\}$ and $B = \{b_0, b_1, \dots, b_{n-1}\}$ and $E = \{(a_i, b_j) : i \leq j\}$

Example 1.1

Consider the half-graph of length 3 (see Figure 1). Then we get that

$$A = \{a_0, a_1, a_2\}, \quad B = \{b_0, b_1, b_2\}$$

$$E = \{(a_0, b_0), (a_0, b_1), (a_0, b_2), (a_1, b_1), (a_1, b_2), (a_2, b_2)\}$$

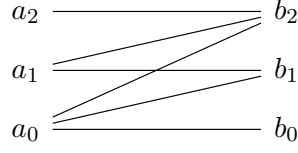


Figure 1: Half-graph of length 3 (H_3)

It turns out that half-graphs cannot be partitioned equitably without having irregular pairs. Furthermore, the authors in [4] showed that the existence of half-graphs as a subgraph of G characterize how many irregular pairs are needed in the equipartition of G and give a better bound in the number of subsets in said equipartition with respect to ε than the one that had already been demonstrated by Gowers in [2]. With the following definition, these results boil down to the the main theorem of this paper:

Definition 1.2 (k -edge stable graphs)

Given $k \in \mathbb{N}$, we say a graph G is **k -edge stable** if it contains no half-graph of length k .

Theorem 1.2 (Stable Regularity Lemma [4])

For each $\varepsilon > 0$ and $k \in \mathbb{N}$ there exists $N = N(\varepsilon, k)$ such that for any sufficiently large finite k -edge stable graph G , for some $l \leq N$, G can be partitioned into disjoint pieces $A_1, \dots, A_l$ such that:

- $||A_i| - |A_j|| \leq 1$
- All pairs (A_i, A_j) are ε -regular, moreover they have density either $> 1 - \varepsilon$ or $< \varepsilon$
- $N < \left(\frac{4}{\varepsilon}\right)^{2^{k+3}-4}$

2 Preliminaries

In order to prove Theorem 1.2 we need to borrow some concepts from model theory. This section introduces such concepts. From here on we will refer to $A \subseteq G$ as the subgraph of $G = (V, E)$ induced by the set of vertices $A \subseteq V$ and say $b \in G$ for any vertex $b \in V$.

2.1 ε -good subset

In this subsection we introduce the first concept from model theory that we will use: ε -good subsets.

Definition 2.1 (ε -good subset)

Call $A \subseteq G$ **ε -good** if for every $b \in G$ one of the following two holds:

- $|\{a \in A : (a, b) \in E\}| < \varepsilon|A|$
- $|\{a \in A : (a, b) \notin E\}| < \varepsilon|A|$

What it means for a subset A to be ε -good is that for every node b in G all but at most a ε -fraction of the nodes in A agree on either having an edge with b or not having an edge

with b . Something important to note is that for any $a \in G$ we get that $A = \{a\}$ is ε -good since for any $b \in G$ either $(a, b) \in E$ in which case $|\{a \in A : (a, b) \notin E\}| = 0 < \varepsilon|A| = \varepsilon$, or $(a, b) \notin E$ in which case $|\{a \in A : (a, b) \in E\}| = 0 < \varepsilon|A| = \varepsilon$. Recurrently in this paper we will use this interpretation of A having a majority vote and agreeing on either connecting to b or not. We formalize it as follows:

Definition 2.2 (Opinion of ε -good subsets)

Let $A \subseteq G$ be a ε -good subset and b a node in G . We say the **opinion** of A with respect to b is $\mathbf{t}(b, A)$ where

$$\mathbf{t}(b, A) = \begin{cases} 0 & \text{if } |\{a \in A : (a, b) \in E\}| < \varepsilon|A| \\ 1 & \text{if } |\{a \in A : (a, b) \notin E\}| < \varepsilon|A| \end{cases}$$

2.2 ε -excellent subgraph

Seeing that ε -good means that the elements in A behave in a regular-like way with respect to every node in G it is just natural to ask ourselves if it is possible to extend this definition to compare A with other subsets, and from there go into ε -regular pairs. Luckily, there exists said extension: it is called ε -excellent subsets.

Definition 2.3 (ε -excellent subgraph)

Call $A \subseteq G$ ε -excellent if for every $B \subseteq G$ such that B is ε -good one of the following two holds:

- $|\{a \in A : \mathbf{t}(a, B) = 1\}| < \varepsilon|A|$
- $|\{a \in A : \mathbf{t}(a, B) = 0\}| < \varepsilon|A|$

In other words, what it means for a subset A to be ε -excellent is that for every subset B that has a clear opinion over all the nodes in G that opinion is the same with respect to all but at most a ε -fraction of the nodes in A .

To see a direct connection between being ε -excellent and being ε -good other than the definition notice that if A is ε -excellent then it also is ε -good. This is true because every singleton induces a ε -good subset, and the opinion of a singleton with respect to a node is exactly the same as if there is an edge between each other, thus A is good with respect all singletons i.e. it is ε -good.

2.3 Full special trees

In this subsection we introduce the third concept from model theory which is rather different from the two introduced previously: Full special trees.

Definition 2.4 (Full special trees)

Let 2^m represent the set of binary strings of length m and $2^{<m} = \bigcup_{i=0}^{m-1} 2^i$. For $\rho \in 2^m$ let $\rho[k]$ be the first k most significant bits of ρ and $\rho[k]$ be the $(k+1)^{th}$ most significant bit of ρ (we use 0-indexing). A **full special tree** of height n in a graph $G = (V, E)$ is a configuration consisting of two sets of vertices $\{b_\rho : \rho \in 2^{<n}\}$, called branches, and $\{a_\eta : \eta \in 2^n\}$, called leaves, such that if $\rho \in 2^m$ then for all η such that $\rho = \eta[:m]$ it holds that

$$(b_\rho, a_\eta) \in E \iff \eta[m] = 1,$$

no further constraint on the edges is given.

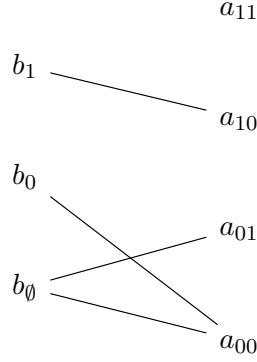


Figure 2: Full special tree of height 2

Example 2.1 ([5])

Consider a special tree of height 2 (see Figure 2) with nodes $b_\emptyset, b_0, b_1$ and leaves $a_{00}, a_{01}, a_{10}, a_{11}$. Then the following must hold

$$\begin{aligned} (b_\emptyset, a_{00}), (b_\emptyset, a_{01}), (b_0, a_{00}), (b_1, a_{10}) &\notin E \\ (b_\emptyset, a_{10}), (b_\emptyset, a_{11}), (b_0, a_{01}), (b_1, a_{11}) &\in E \end{aligned}$$

This is the hardest concept to tie to ε -regularity from the three presented. However, we will show that there is a strong relation between the existence of full special trees and half-graphs as subgraphs of G . The result is portrayed in Theorem 2.1 given the following definition:

Definition 2.5 (Tree bound)

Let G be a k -edge stable graph. Call $t = t(k)$ the tree bound of G and let it be the strict upper bound on the height of a full special tree in G .

Theorem 2.1

The following hold:

- (1) If G contains no full special trees of height n , then G is 2^n -edge stable.
- (2) If the graph G is k -edge stable, then there is no full special tree in G of height $2^{k+2} - 2$, i.e. $t(k) \leq 2^{k+2} - 2$

Proof. Let us first demonstrate (1). It is sufficient to show that if there is a half-graph of length 2^n then there is a full special tree of height n . Let $H_{2^n} = (A_{2^n} \cup B_{A_{2^n}}, E)$ be a half graph of length 2^n , we will show that there is a full special tree of height n as a subgraph of H_{2^n} . Let $b_i \in B$ be relabeled as $b'_{bin_n(i)}$ where $bin_n(i)$ is the binary expansion of i into n bits (possible since $0 \leq i < 2^n$). Then for each $\rho \in 2^{<n}$ relabel $a_i \in A$ to a'_ρ where $bin_n(i) = \rho || 100 \dots 0$. Notice that under this relabeling we get that for all ρ, η, i, j such that $\rho = \eta[:m]$ and $a_i = a'_\rho$ and $b_j = b'_\eta$ it holds that:

- If $\eta[m] = 1$ then $i \leq j$ since i is the smallest number such that $bin_n(i)[:m+1] = \rho || 1$
- If $\eta[m] = 0$ then $i > j$ since $bin_n(i)[:m+1] = \rho || 1$ and $bin_n(j)[:m+1] = \rho || 0$.

Therefore if $\rho \in 2^m$ then for all η such that $\rho = \eta[:m]$ we get $(b_\rho, a_\eta) \in E \iff \eta[m] = 1$, i.e., this relabeling implies that a full special tree of height n is a subgraph of H_{2^n} .

For the demonstration of (2) refer to lemma 6.7.9 in [3]. $\square$

3 Building back to ε -regularity

Now that we have introduced some tools from model theory and given some intuition on how they are related to ε -regularity it is time to move towards the goal of demonstrating Theorem 1.2.

3.1 From ε -excellence to ε -regularity

We have hinted at the idea that ε -excellent subsets present regular-like behavior; in fact, two ε -excellent subsets behave uniformly between them and make a ζ -regular pair for $\zeta = \zeta(\varepsilon)$. Formally we make the following claim:

Claim 3.1

Suppose $A \subseteq G$, $B \subseteq G$ and $\varepsilon < \frac{1}{2}$ such that A, B are ε -excellent. Then:

- (a) The pair (A, B) is ε -uniform, meaning that there exists a $\mathbf{t}(A, B) \in \{0, 1\}$ such that for all but at most $\varepsilon|A|$ elements $a \in A$ and $\varepsilon|B|$ elements $b \in B$ it holds that $(a, b) \in E \iff \mathbf{t}(A, B) = 1$.
- (b) For $\zeta = \sqrt{2\varepsilon}$, the pair (A, B) is ζ -regular. Moreover $d(A, B) > 1 - \zeta$ or $d(A, B) < \zeta$.

Proof.

- (a) This result is straight-forward from the definition of ε -excellent subsets. Since A and B are ε -excellent then they are ε -good which means there exists a majority opinion of A with respect to B and a majority opinion of B with respect to A , furthermore, they must be the same since both of these are opinions with respect to the existence of edges between A and B . From the definition of ε -excellent subsets we also get that there are at most $\varepsilon|A|$ nodes that disagree in A and $\varepsilon|B|$ nodes that disagree in B . Since this is an opinion between A and B it is natural to extend the notation $\mathbf{t}(A, B)$.
- (b) Let $A' \subseteq A$ and $B' \subseteq B$ such that $|A'| \geq \zeta|A|$ and $|B'| \geq \zeta|B|$. Let

$$\mathcal{W}_a = \begin{cases} \{b \in B : (a, b) \notin E\} & \text{if } \mathbf{t}(A, B) = 1 \\ \{b \in B : (a, b) \in E\} & \text{if } \mathbf{t}(A, B) = 0 \end{cases}$$

be the set of elements $b \in B$ whose opinion with respect to $a \in A$ does not agree with the general opinion between A and B . Let $\mathcal{U} = \{a \in A : |\mathcal{W}_a| > \varepsilon|A|\}$ be the set of elements $a \in A$ such that at least $\varepsilon|A|$ of its neighbors in B disagree with $\mathbf{t}(A, B)$. Now, A being ε -excellent implies that $|\mathcal{U}| < \varepsilon|A|$, and B being ε -good implies that if $a \in A \setminus \mathcal{U}$ we know $|\mathcal{W}_a| < \varepsilon|B|$. Now, let

$$Z = \begin{cases} \{(a, b) \in (A \times B) : (a, b) \notin E\} & \text{if } \mathbf{t}(A, B) = 1 \\ \{(a, b) \in (A \times B) : (a, b) \in E\} & \text{if } \mathbf{t}(A, B) = 0 \end{cases}$$

$$Z' = \begin{cases} \{(a, b) \in (A' \times B') : (a, b) \notin E\} & \text{if } \mathbf{t}(A, B) = 1 \\ \{(a, b) \in (A' \times B') : (a, b) \in E\} & \text{if } \mathbf{t}(A, B) = 0 \end{cases}$$

be the sets of pairs that do not agree with $\mathbf{t}(A, B)$ in $A \times B$ and $A' \times B'$. Notice then that

$$Z \subseteq \mathcal{U} \times B \cup \bigcup \{(a, b) \in (A \times B) : b \in \mathcal{W}_a, a \in A \setminus \mathcal{U}\}$$

$$Z' \subseteq \mathcal{U} \times B' \cup \bigcup \{(a, b) \in (A' \times B) : b \in \mathcal{W}_a, a \in A \setminus \mathcal{U}\}$$

Therefore we get that

$$\begin{aligned} |Z| &\leq |\mathcal{U}| \cdot |B| + |A| \cdot \max\{|\mathcal{W}_a| : a \in A \setminus \mathcal{U}\} < \varepsilon|A| \cdot |B| + |A| \cdot \varepsilon|B| = 2\varepsilon|A||B| \\ &\implies \frac{|Z|}{|A \times B|} < 2\varepsilon \end{aligned} \tag{1}$$

and, similarly,

$$\begin{aligned} |Z'| &\leq |\mathcal{U}| \cdot |B'| + |A'| \cdot \max\{|\mathcal{W}_a| : a \in A \setminus \mathcal{U}\} \\ &\implies |Z'| < \varepsilon|A| \cdot \zeta|B| + \zeta|A| \cdot \varepsilon|B| = 2\zeta\varepsilon|A||B| \leq \frac{2\varepsilon}{\zeta}|A'||B'| \\ &\implies \frac{|Z'|}{|A' \times B'|} < \frac{2\varepsilon}{\zeta} \end{aligned} \tag{2}$$

Notice that if $\mathbf{t}(A, B) = 1$ inequality (1) implies that $d(A, B) > 1 - 2\varepsilon$ while inequality (2) implies that $d(A', B') > 1 - \frac{2\varepsilon}{\zeta}$; similarly if $\mathbf{t}(A, B) = 0$ inequality (1) implies that $d(A, B) < 2\varepsilon$ while inequality (2) implies that $d(A', B') < \frac{2\varepsilon}{\zeta}$. Therefore $|d(A, B) - d(A', B')| < \frac{2\varepsilon}{\zeta}$, hence, for $\zeta = \sqrt{2\varepsilon}$ we get $\zeta = \frac{2\varepsilon}{\zeta}$ which implies that if $|A'| \geq \zeta|A|$ and $|B'| \geq \zeta|B|$ then $|d(A, B) - d(A', B')| < \zeta$ i.e., (A, B) is a ζ -regular pair, as we wanted. $\square$

3.2 From half-graphs and full special trees to ε -excellent partitions

It is not obvious that ε -excellent sets exist in any given graph. In this subsection we show that it is sufficient for a graph to be k -edge stable for ε -excellent subsets to exist.

Claim 3.2

Suppose G is a k -edge stable graph, $t = t(k)$ is the tree bound, and $\varepsilon < \frac{1}{2t}$. Then for every A with $|A| \geq \frac{1}{\varepsilon^t}$ there exists a ε -excellent subset $A' \subseteq A$ with $|A'| \geq \varepsilon^{t-1}|A|$.

Proof. ([5]) Lets proceed by contradiction. Lets perform the following inductive construction of the sets $\langle A_\eta : \eta \in 2^m \rangle$ and $\langle B_\eta : \eta \in 2^m \rangle$ for $0 \leq m \leq t$ such that B_ρ witnesses that A_ρ is not ε -excellent:

- For $m = 0$ we get $A_\emptyset = A$ and let $B_\emptyset$ witness that ε -excellent.
- For $0 < m \leq t$ and all η that have length $m - 1$ we construct

- $A_{\eta||i} = \{a \in A_\eta : \mathbf{t}(a, B_\eta) = i\}$, for $i \in \{0, 1\}$
- $B_{\eta||i}$ such that $B_{\eta||i}$ witnesses that $A_{\eta||i}$ is not ε -excellent, for $i \in \{0, 1\}$

Notice that, since B_η is a witness that A_η is not ε -excellent, it must be that both $A_{\eta||0}$ and $A_{\eta||1}$ have size at least $\varepsilon|A_\eta|$. Furthermore, notice that if there exists ρ of length less than t such that A_ρ is ε -excellent (i.e. we cannot find B_ρ) then $|A_\rho| \geq \varepsilon^{t-1}|A|$ and we are done. Therefore, lets suppose for the sake of contradiction that we can in fact perform the construction above for all $0 \leq m \leq t$. We'll show that this implies contradiction by constructing a full special tree of height t .

We first choose the leaves: for each $\eta \in 2^t$ let a_η be an arbitrary element of A_η which exists since $|A_\eta| \geq \varepsilon^t |A| \geq 1$ since $|A| \geq \frac{1}{\varepsilon^t}$. Now we choose the branches: for each $\rho \in 2^m$ let U_ρ be the set of bad (i.e. not good) nodes in B_ρ with respect to a_η for the A_η 's that are built from B_ρ ; formally, for a fixed $\rho \in 2^m$ and all $\eta \in 2^t$ such that $\eta[:m] = \rho$ let

$$U_\eta = \begin{cases} \{b \in B_\rho : (a_\eta, b) \notin E\} & \text{if } \mathbf{t}(a_\eta, B_\rho) = 1 \\ \{b \in B_\rho : (a_\eta, b) \in E\} & \text{if } \mathbf{t}(a_\eta, B_\rho) = 0 \end{cases}$$

$$U_\rho = \bigcup \{U_\eta : \eta \in 2^t, \eta[:m] = \rho\}$$

Notice that $|U_\eta| < \varepsilon |B_\rho|$ since B_ρ is ε -good as it is a witness of ε -excellence, therefore $|U_\rho| \leq 2^t \varepsilon |B_\rho| < |B_\rho|$ since $\varepsilon < \frac{1}{2^t}$. Therefore $B_\rho \setminus U_\rho \neq \emptyset$, and we choose branch b_ρ to be an element of $B_\rho \setminus U_\rho$, thus b_ρ is good with respect to all the leaves a_η such that $\eta[:m] = \rho$ i.e. $\langle b_\rho \rangle$ and $\langle a_\eta \rangle$ make a full special tree of height t . $\square$

Now that we know we can get ε -excellent subsets for large enough k -edge stable graphs then we can use claim 3.2 inductively to construct a partition into ε -excellent subsets and a set of leftover vertices that we would like to distribute without causing much trouble.

3.3 From existence of ε -excellent subsets to excellent equipartitions

There is still a problem we have to address even after being able to construct a partition of ε -excellent subsets: the partition we are constructing is not necessarily (and rather unlikely to be) an equipartition. To solve this we will modify Claim 3.1 using the following definition:

Definition 3.1 (Size sequence)

Call the sequence $s_0, s_1, \dots, s_{t-1}$ a size sequence for ε when:

- (a) $\varepsilon s_l \geq s_{l+1}$ for $l < t-1$
- (b) $s_{t-1} > t$
- (c) s_{t-1} divides all elements s_i of the sequence.

Claim 3.3

Suppose G is a k -edge stable graph, $t = t(k)$ its tree bound, and $\varepsilon < \frac{1}{2^t}$. Let $s_0, s_1, \dots, s_{t-1}$ be a size sequence for ε . Then for every $A \subseteq G$ with $|A| \geq \max\{s_0, \frac{1}{\varepsilon^t}\}$ there exists a ε -excellent subset $A' \subseteq A$ with $|A'| = s_l$ for some $l = 0, \dots, t-1$

Proof. Since this is a modification to Claim 3.2 it naturally follows the same proof with the following modifications:

- First, for $A_\emptyset$ do not choose A but an arbitrary subset of size s_0 .
- From there on, we inductively assume that at each stage $m-1$ the set A_η with η of length $m-1$ has exactly s_{m-1} elements
- For stage m both $A_{\eta||0}$ and $A_{\eta||1}$ have size at least $\varepsilon s_{m-1} \geq s_m$ and we drop arbitrary vertices to obtain size exactly s_m if necessary.

$\square$

Notice that condition (c) of Definition 3.1 suggests that what we really are aiming for with Claim 3.3 is to get an equipartition into sets of size s_{t-1} . However we still need one more result to do so.

Claim 3.4 (Adapted from [1] proposition 4.5)

Let $\zeta > \varepsilon > 0$ and $r, k \geq 1$. Let $A \subseteq G$ be a ε -excellent subset of size $n = r \cdot c$ for some natural c , of a k -edge stable graph G with tree bound $t(k)$. Then there exists a constant $N = N(\zeta, \varepsilon, r, k)$ such that if $n \geq N$ then A can be equipartitioned into c ζ -excellent subsets. Moreover, N is the smallest value such that

$$r \log r + 2 < 2\zeta^2 N - r2^{t(k)+1} \log N.$$

4 Stable Regularity Lemma

We are now ready to prove Theorem 1.2. To do this, we will first show a slightly more specific version of the theorem which explicitly mentions most of the concepts introduced in the paper so far.

Theorem 4.1 (Stable Regularity Lemma - Version with excellence [5])

For each $k \geq 1$ and $\varepsilon > 0$, there exists $N = N(k, \varepsilon)$ such that if G is any sufficiently large k -stable graph there is a partition of G into disjoint sets $A_0, A_1, \dots, A_{l-1}$ with $l \leq N$ such that:

- $||A_i| - |A_j|| \leq 1$ for all i, j
- A_i is ε -excellent for every i
- (A_i, A_j) is a $\sqrt{2\varepsilon}$ -regular pair for all i, j
- if $\varepsilon < \frac{1}{2^t}$, then $N \leq 2 \left(\frac{8}{\varepsilon}\right)^t$ where $t = t(k)$ is the tree bound of G .

Proof. Let n be the size of G and let $q = \lceil \frac{4}{\varepsilon} \rceil$. Let $c \in \mathbb{N}$ be the maximum natural number such that $q^{t-1}c \leq \frac{\varepsilon n}{8}$. Define the sequence $\{s_i : s_i = q^{t-1-i}c, 0 \leq i \leq t-1\}$. Notice that $s_{t-1} | s_i$ for all i , that $\varepsilon s_i > q^{-1}s_i = s_{i+1}$, and that for n sufficiently large we have $s_{t-1} = c > t$, therefore $\{s_i\}$ is a size sequence.

We now want to apply Claim 3.3. Notice that to get subsets that are ζ -excellent we require the size of the set we are working with to be at least $\max\{s_0, \frac{1}{\zeta}\} = s_0$ for n sufficiently large. Thus, for n sufficiently large we can apply Claim 3.3 repeatedly to construct a partition $\{B_j\} \cup \{B\}$ of G into $\frac{\varepsilon}{4}$ -excellent subsets, each of size s_l for some $l \in \{0, \dots, t-1\}$ plus a remainder set B such that $|B| < s_0 = q^{t-1}c \leq \frac{\varepsilon n}{8}$.

We now want to apply Claim 3.4. Notice that for n sufficiently large we can ensure that the size of B_i is large enough to apply Claim 3.4 to B_i for all i . This way we can achieve a new partition $\{B'_j\} \cup \{B\}$ such that $|B'_i| = s_{t-1}$ and B'_i is $\frac{\varepsilon}{3}$ -excellent (since $\frac{\varepsilon}{3} > \frac{\varepsilon}{4}$) for all i , where B is still the same remainder set. Now we will distribute this remainder set equally among the B'_i to obtain a new partition $\{A_i\}$ such that $||A_i| - |A_j|| \leq 1$ for all i, j . We will show that every A_i in this equipartition is ε -excellent.

Let $A'_i = A_i \setminus B'_i \in B$ be the set of elements added to B'_i to create A_i . Now notice that B'_i being $\frac{\varepsilon}{3}$ -excellent implies that, even in the worst case where all the elements in A'_i disagree on the opinion of some good subset with respect to B'_i , we at least get that $A_i = A'_i \cup B'_i$ is $\frac{\frac{\varepsilon}{3}|B'_i| + |A'_i|}{|B'_i| + |A'_i|}$ -excellent. It is therefore enough to show that $|A'_i| \leq \frac{2\varepsilon}{3}|B'_i|$.

Let $\gamma = |\{A_i\}|$ be the number of sets in the equipartition. Then notice that

$$\frac{n}{s_{t-1}} \geq \gamma > \frac{n - s_0}{s_{t-1}}$$

since $|B_i| = s_{t-1}$ and $|B| < s_0$. Hence

$$|A'_i| - 1 \leq \left\lfloor \frac{|B|}{\gamma} \right\rfloor \leq \left\lfloor \frac{s_0 - 1}{\gamma} \right\rfloor \leq \frac{s_0 - 1}{\gamma} < (s_0 - 1) \left(\frac{n - s_0}{s_{t-1}} \right)^{-1}$$

and recall that $|B'_i| = s_{t-1}$, thus

$$\frac{|A'_i|}{|B'_i|} - \frac{1}{s_{t-1}} = \frac{|A'_i| - 1}{|B'_i|} < \left(\frac{s_0 - 1}{s_{t-1}} \right) \left(\frac{n - s_0}{s_{t-1}} \right)^{-1} = \frac{s_0 - 1}{n - s_0}$$

however notice that

$$\begin{aligned} s_0 \leq \frac{\varepsilon n}{8} = \frac{\frac{\varepsilon}{4}n}{2} < \frac{\frac{\varepsilon}{3}n + 1}{1 + \frac{\varepsilon}{3}} &\implies s_0 - 1 < \frac{\varepsilon}{3}(n - s_0) \implies \frac{s_0 - 1}{n - s_0} < \frac{\varepsilon}{3} \\ &\implies \frac{|A'_i|}{|B'_i|} - \frac{1}{s_{t-1}} < \frac{\varepsilon}{3} \end{aligned}$$

on the other hand we know that for n large enough $s_{t-1} > \frac{3}{\varepsilon}$, hence $\frac{\varepsilon}{3} > \frac{1}{s_{t-1}}$, therefore

$$\frac{|A'_i|}{|B'_i|} - \frac{1}{s_{t-1}} < \frac{\varepsilon}{3} \implies \frac{|A'_i|}{|B'_i|} < \frac{2\varepsilon}{3} \implies |A'_i| \leq \frac{2\varepsilon}{3}|B'_i|$$

as we wanted. Since A_i is ε -excellent for every i , using Claim 3.1 we get that (A_i, A_j) is a $\sqrt{2\varepsilon}$ -regular pair for all i, j . Lastly we want to bound the number of sets in the equipartition, i.e. γ . Notice that from the way we chose c we know that $\frac{\varepsilon n}{8} - q^{t-1} < q^{t-1}c$, then $\frac{\varepsilon n}{8q^{t-1}} - 1 < c$, thus $\frac{\varepsilon n}{16q^{t-1}} < c$. Notice also that $q = \lceil \frac{4}{\varepsilon} \rceil \leq \frac{8}{\varepsilon}$, therefore we can bound γ by

$$\gamma \leq \frac{n}{c} \leq n \left(\frac{\varepsilon n}{16q^{t-1}} \right)^{-1} = \frac{16q^{t-1}}{\varepsilon} \leq \frac{16 \left(\frac{8}{\varepsilon} \right)^{t-1}}{\varepsilon} = 2 \left(\frac{8}{\varepsilon} \right)^t$$

□

With Theorem 4.1 demonstrated we can finally demonstrate Theorem 1.2

Proof of Theorem 1.2 Apply Theorem 4.1 with k and $\frac{\varepsilon^2}{2}$ (so that $\sqrt{2\frac{\varepsilon^2}{2}} = \varepsilon$). Since the statement of Theorem 1.2 does not mention t simply use Theorem 2.1 to replace it by the upper bound $2^{k+2} - 2$.

□

5 Conclusion

We have shown that the existence of irregular pairs for a graph G in Szemerédi's regularity lemma depends on the existence of special structures called half-graphs as subgraphs of G ; if there is a fixed bound on the size of such structure as a subgraph for an arbitrarily large graph G then we can give an equipartition that contains no irregular pairs whatsoever. Furthermore, the result obtained from the nonexistence of said structures is even stronger as it gives a much better bound for the number of sets in the equipartition of G than that given in the original demonstration by Szemerédi and the lower bound further found by Gowers for general graphs.

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